LED lighthouse

Motion + darkness -> about 60 seconds of five sequential white LEDs -> all LEDs off

Build status: datasheet-based prototype, not electrically simulated or tested on hardware. Use this guide to assemble and measure one stage at a time. This is a connection schematic, not a PCB or breadboard hole layout.

1 DETECT

PIR senses a new motion edge.
LDR permits operation only in darkness.

2 RUN

U1 enables the LED supply for about one minute.
Extra motion does not restart the timer.

3 SEQUENCE

U2 + U4 select LED1 to LED5 repeatedly.
Idle resets the sequence to LED1.

Power arrangement

Protected 1S battery

3.7V nominal, 4.2V charge chemistry.
Known charge-current rating required.

LaskaKit LA122076

Battery terminals verified on actual board.
USB-C for charging; 5V boost output.

S1 -> J1 -> circuit

Switch the verified +5V output.
GND is common across all stages.

Power first: use a current-limited regulated 5V supply for initial checks, or a verified USB 5V source. Do not connect an unidentified recycled cell.

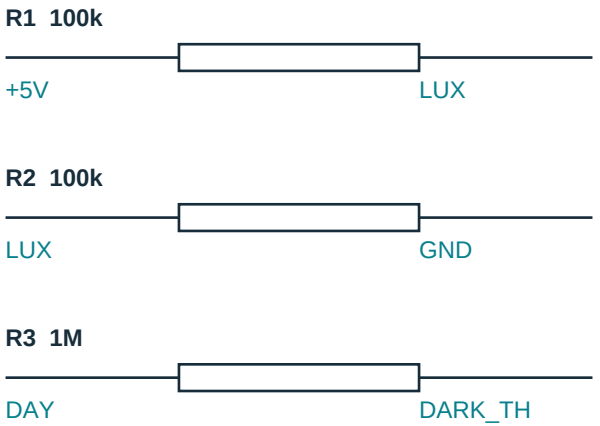
Board terminals: J1 and J2 use signal names, not an assumed physical pin order. Match the actual PCB labels. If a module is unlabelled, obtain its pinout before connecting it.

Behaviour: daylight during a run cancels it. A PIR startup pulse can activate the lighthouse while the sensor settles; allow at least 60 seconds after power-on before testing.

Reading: matching teal net names on different sheets are the same wire. NC means leave open. Component pin numbers are top view, notch upward.

01 / Darkness and reset

U3A permits night operation; U3B resets the LED sequence whenever RUN is low



LDR1: LUX to GND. C14: LUX to GND. Neither has polarity.

Network	Exact connections
Darkness setting	R4: +5V to REF_TOP, 47k. R5: REF_TOP to GND, 47k. P1 A=REF_TOP, W=DARK_TH, B=GND (100k).
Daylight output	R38: +5V to DAY, 100k. R39: DAY to Q12 base, 47k. Q12 C=ARM, E=GND. R6: +5V to ARM, 10k. C6: ARM (+) to GND (-), 100u.
Idle detection	R7: RUN to RUN_HALF, 100k. R8: RUN_HALF to GND, 100k. R9: +5V to FIX_REF, 47k. R10: FIX_REF to GND, 15k. R11: +5V to IDLE, 10k.

Inputs remain below 3.5V at a 5V supply. DAY is high in light, low in darkness. Q12 pulls ARM low in light. In darkness C6 charges and releases U1 RESET. P1 clockwise direction depends on which outer terminal is A; start at mid travel.

02 / Motion pulse and lockout

Buffered edge trigger: no capacitive signal is applied directly to a TLC555 input

Signal path

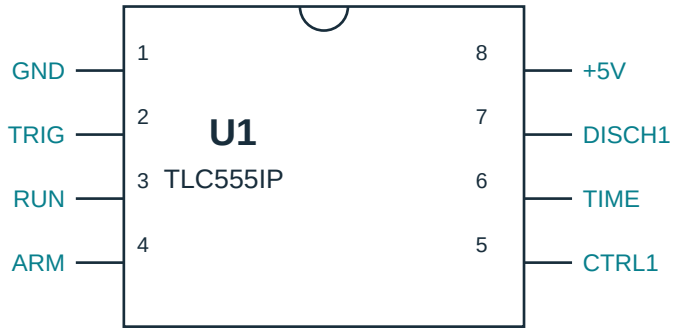
PIR_OUT -> R12 -> Q1 -> INV -> R14 -> Q2 -> MOTION_BUF -> C11 -> EDGE -> R16 -> Q3 base. A rising PIR output produces a short LOW pulse at TRIG.

Part / network	Connection
Q1 input inverter	R12=47k from PIR_OUT to Q1.B; Q1.E=GND; Q1.C=INV; R13=47k from +5V to INV.
Q2 motion buffer	R14=47k from INV to Q2.B; Q2.E=GND; Q2.C=MOTION_BUF; R15=47k from +5V to MOTION_BUF.
Q3 pulse switch	C11=100n between MOTION_BUF and EDGE. R16=10k from EDGE to Q3.B. R17=100k from Q3.B to GND. Q3.E=GND; Q3.C=TRIG; R18=10k from +5V to TRIG.
D1 base clamp	D1 anode=GND; cathode (band)=Q3.B. It limits reverse base voltage on the falling edge.
Q4 busy lockout	R19=47k from RUN to Q4.B; Q4.E=GND; Q4.C=Q3.B. C15=10u, (+)=Q4.B, (-)=GND. RUN disables Q3; C15 holds lockout briefly after timeout to reject an overlapping edge.

The PIR can stay high during continued movement. C11 responds to edges, so a held-high PIR output does not hold TRIG low. After timeout and a short lockout tail, the next LOW-to-HIGH PIR transition can start a new run.

03 / Duration and LED speed

Both ICs are TLC555IP, DIP8; use the numbered pins, not the text position on the package



Circuit	Timing network and settings
U1 minute timer	+5V -> R20 220k -> P3 A. Strap P3 W and B together at TIME. C7 100u: (+)=TIME, (-)=GND. U1 pin 6=TIME. TIME -> R40 100 ohm -> U1 pin 7 (DISCH1). C8 10n from pin 5 to GND.
U2 LED oscillator	+5V -> R21 10k -> DISCH2 (pin 7) -> R22 10k -> P2 A. Strap P2 W and B at OSC. U2 pins 2 and 6=OSC. C10 10u: (+)=OSC, (-)=GND. C9 10n from pin 5 to GND.
Local supply bypass	C1 100n directly between U1 pins 8 and 1. C2 100n directly between U2 pins 8 and 1.

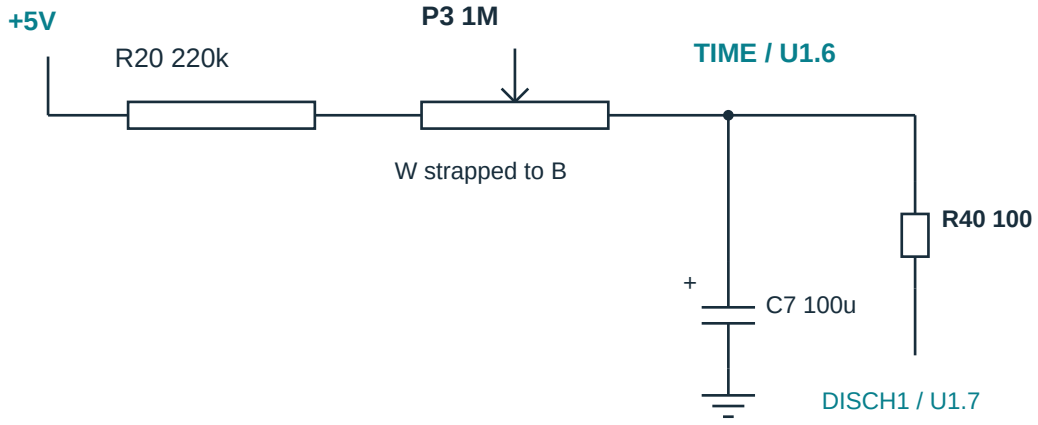
Duration: t approximately 1.1 x (220k + P3) x 100u: nominally 24 to 134 seconds. Start P3 near 330k for about 60 seconds. Measure and adjust; capacitor tolerance and leakage can substantially change the result.

Step interval: T approximately 0.693 x [10k + 2 x (10k + P2)] x 10u: about 0.21 to 1.59 seconds per LED. Start P2 near 35k for about 0.69 seconds per step. The first LED dwell after starting is different from steady-state timing.

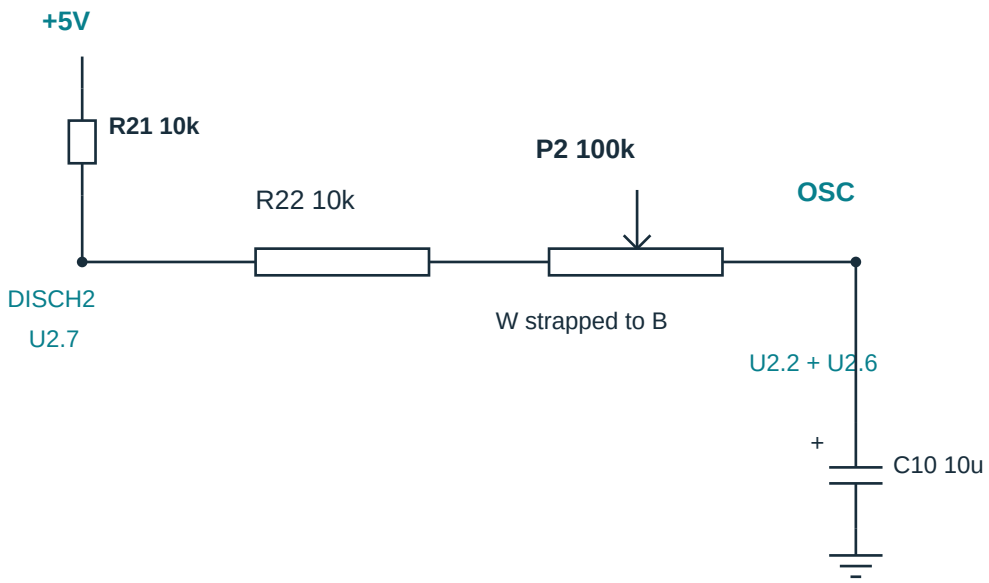
03A / Timing network schematics

These drawings expand the TIME, DISCH1, DISCH2 and OSC nets from sheet 03

U1 / about one minute



U2 / LED stepping speed

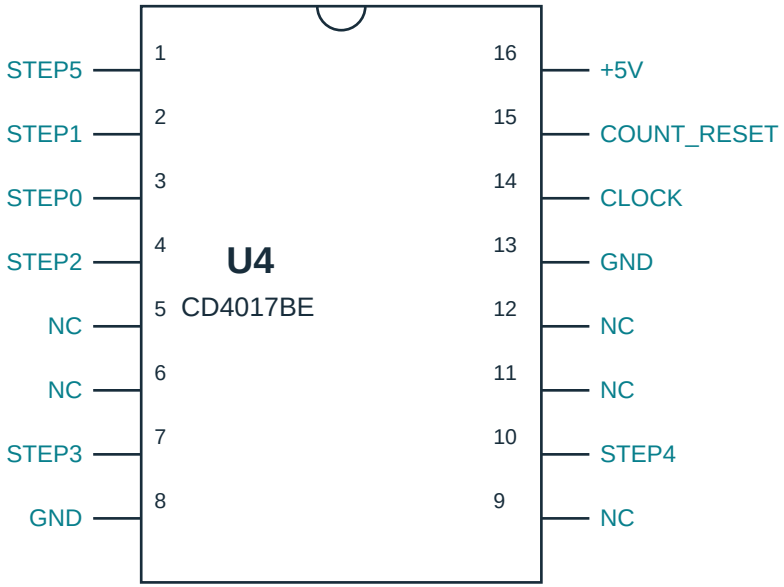


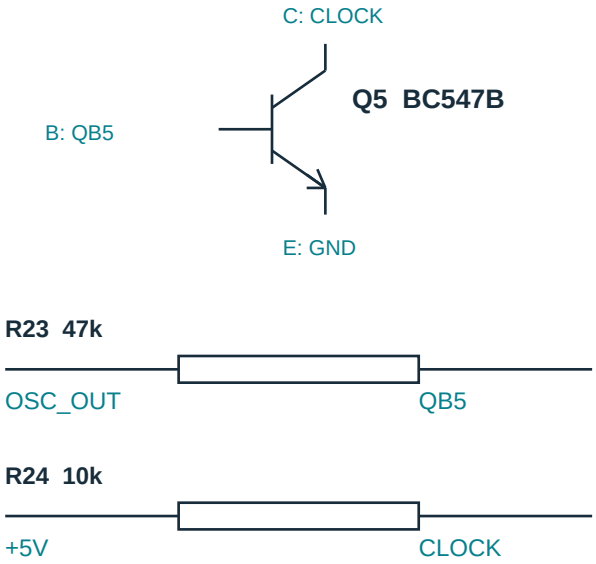
How to wire the trim pots: choose one outer terminal as A. Connect the wiper W to the other outer terminal B. The resistance from A to W+B is adjustable from nearly zero to the pot's rated resistance. Verify with a meter while unpowered.

Capacitor polarity: C7 positive to TIME, C10 positive to OSC; both negative leads to GND. R40 prevents a large current surge when U1 discharges C7. Do not join DISCH1 directly to TIME.

04 / Five-step counter

Clock inversion preserves LED1 at the start; diode OR combines idle reset and modulo-five reset





Connection	Details
Clock	U2.3=OSC_OUT -> R23 -> Q5.B; Q5.E=GND; Q5.C=CLOCK. R24 pulls CLOCK up to +5V. CLOCK -> U4.14.
Reset OR	D2 A=IDLE, K=COUNT_RESET. D3 A=STEP5 (U4.1), K=COUNT_RESET. R25=100k and C12=10n each from COUNT_RESET to GND. COUNT_RESET -> U4.15.
Always connected	U4.16=+5V; U4.8=GND; U4.13=GND. C4=100n between pins 16 and 8. Pins 5, 6, 9, 11 and 12 stay unconnected.

Sequence: U4.3 -> U4.2 -> U4.4 -> U4.7 -> U4.10 -> repeat. STEP5 at pin 1 is a reset signal, not a sixth LED.

05 / Switched LED supply

Q6 is an emitter follower, not a grounded-emitter switch; connect its three terminals exactly as shown

Q6 collector goes to +5V; its emitter provides LED_POWER (typically about 4V during RUN, near 0V when idle). This deliberate voltage drop leaves ample headroom for low-current white LEDs while ensuring that they all turn off.

Repeat this branch five times

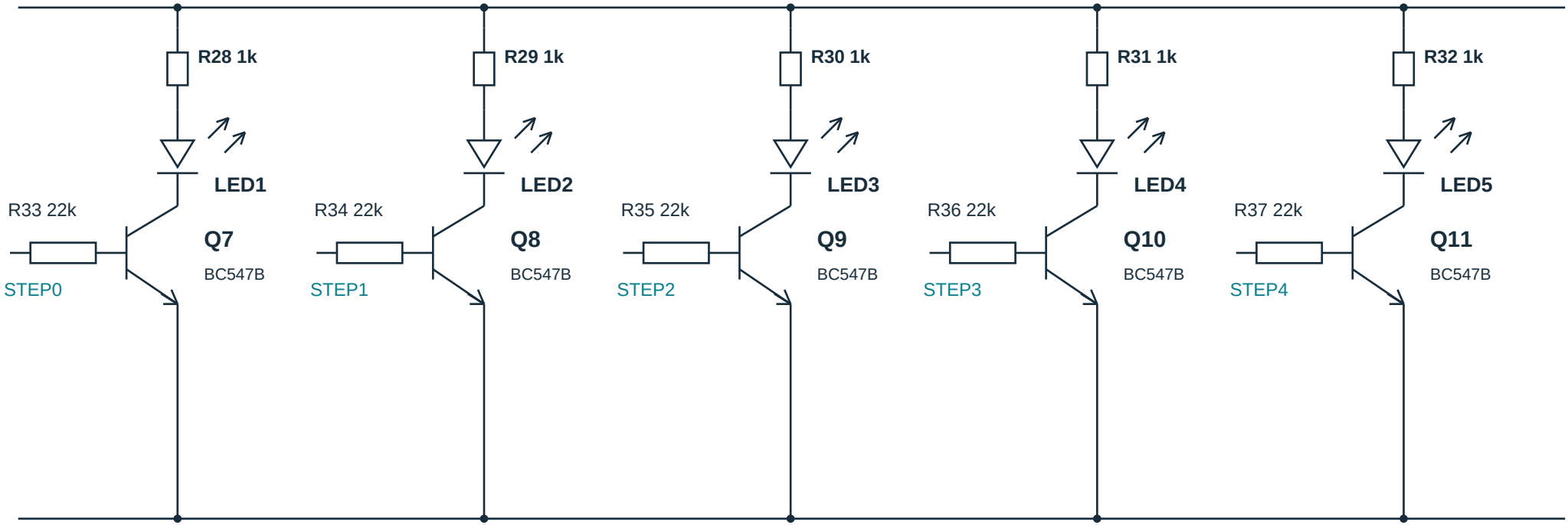
LED position	Counter output -> base resistor -> transistor	Light-current path
LED1	U4.3 (STEP0) -> R33 22k -> Q7.B	LED_POWER -> R28 1k -> LED1.A; LED1.K -> Q7.C; Q7.E -> GND
LED2	U4.2 (STEP1) -> R34 22k -> Q8.B	LED_POWER -> R29 1k -> LED2.A; LED2.K -> Q8.C; Q8.E -> GND
LED3	U4.4 (STEP2) -> R35 22k -> Q9.B	LED_POWER -> R30 1k -> LED3.A; LED3.K -> Q9.C; Q9.E -> GND
LED4	U4.7 (STEP3) -> R36 22k -> Q10.B	LED_POWER -> R31 1k -> LED4.A; LED4.K -> Q10.C; Q10.E -> GND
LED5	U4.10 (STEP4) -> R37 22k -> Q11.B	LED_POWER -> R32 1k -> LED5.A; LED5.K -> Q11.C; Q11.E -> GND

Each LED has its own 1k resistor. Start with these values; expect roughly 0.5-1.5mA for typical white LEDs. Actual LED forward voltage and brightness must be measured. Do not connect LEDs directly to CD4017 outputs or directly to the battery.

05A / Five LED branches

All five LED anodes use the switched LED_POWER rail from Q6, not the permanent +5V rail

LED_POWER from Q6.E



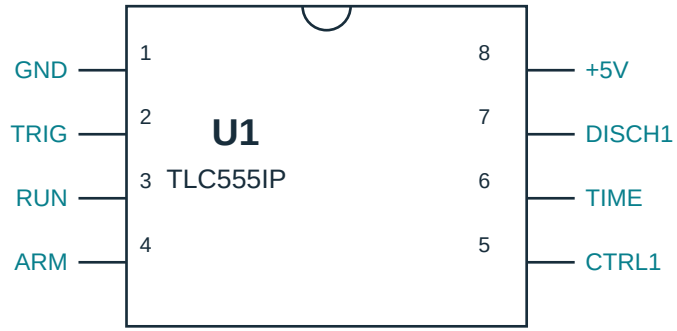
GND

Counter outputs: STEP0=U4 pin 3; STEP1=pin 2; STEP2=pin 4; STEP3=pin 7; STEP4=pin 10. Wire each output to its own 22k base resistor. All five emitters join GND.

LED polarity: A (anode) is above the LED symbol, connected through 1k to LED_POWER. K (cathode, bar in the symbol) connects to the transistor collector. Identify the actual LED polarity before soldering; do not rely only on lead length for used LEDs.

06 / Pin orientation and assembly

All DIP diagrams are TOP VIEW: notch upward, pin 1 at top left; numbering runs counterclockwise



Item	How to identify and connect it
BC547B Q1-Q12	Manufacturer default TO-92 drawing: flat face toward you, leads downward, C-B-E from left to right. Verify the received manufacturer / package against its datasheet; special pinouts and substitutions exist. The circuit netlist names C, B, E rather than assuming generic TO-92 numbering.
1N4148 D1-D3	The painted band marks K (cathode). A is the unbanded end. D1 band to Q3.B; D2 and D3 bands to COUNT_RESET.
Electrolytic capacitors	C5, C6, C7, C10 and C15 are polarized. The body stripe normally marks the negative lead: all negative leads here go to GND. Follow the + / - entries in the parts table.
P1, P2, P3 trim pots	Identify the wiper W from the pot drawing or a resistance measurement. Do not infer pin order from how a pot is drawn here. P1 uses all three terminals; P2/P3 strap W to outer terminal B.
PIR and boost module	Connect by verified PCB signal labels only. No physical pad order has been inferred. PIR OUT must be measured LOW at rest and HIGH on motion. If labels or output polarity differ, resolve that before connecting J2.

Fit sockets first, then resistors/diodes, then capacitors. Insert ICs only after confirming supply polarity. Keep bypass capacitors close to the supply pins and the long TIME wire short and clean.

07 / Parts actually used

Counts below are derived from the complete connection netlist, replacing the earlier shopping estimates

Type / value	Used	References
R: 100k	8	R1, R2, R7, R8, R17, R25, R27, R38
R: 1M	1	R3
R: 47k	10	R4, R5, R9, R12, R13, R14, R15, R19, R23, R39
R: 10k	7	R6, R11, R16, R18, R21, R22, R24
R: 15k	1	R10
R: 220k	1	R20
R: 1k	6	R26, R28, R29, R30, R31, R32
R: 22k	5	R33, R34, R35, R36, R37
R: 100	1	R40
C: 100n	7	C1, C2, C3, C4, C11, C13, C14
C: 100u / 16V	3	C5, C6, C7
C: 10n	3	C8, C9, C12
C: 10u / 16V	2	C10, C15

Also used: 2 x TLC555IP, 1 x TLC372CP, 1 x CD4017BE, **12 x BC547B**, 3 x 1N4148, 2 x 100k trim pots, 1 x 1M trim pot, 5 white LEDs and 1 LDR5516. Sockets: 3 x DIP8 and 1 x DIP16. The transistor design uses all 12 ordered pieces.

Resistors are standard 0.25W parts. Check your resistor assortment against the value/count table before assembly. Series combinations are possible where necessary; record any substitutions in the wiring notes.
Supply capacitors: C3=100n between U3 pins 8 and 4; C5=100u (+ to +5V, - to GND); C13=100n between +5V and GND near the PIR. C1-C4 belong directly at their IC supply pins.

08 / First power-up and adjustment

Complete each stage before moving on; the battery is the last power source to connect

Step	Check / expected result
1. Unpowered	No short between +5V and GND. Verify DIP notch orientation, diode bands, transistor C/B/E, capacitor polarity and all pot wipers. Leave PIR unplugged.
2. Supply	Apply regulated 5V with current limit initially 20mA. Confirm about 4.75-5.25V at every IC. If current limit is reached or a component heats, switch off and inspect. Do not increase the limit to mask a wiring fault.
3. Darkness	Cover LDR1. Adjust P1: ARM should rise near 5V. Illuminate LDR: ARM should go below 0.2V. DAY does not need to reach 5V; it drives a transistor. Check LUX / DARK_TH remain below 3.5V.
4. Manual start	With ARM high and LEDs disconnected, briefly connect TRIG to GND through 1k. RUN should go high, then low once. Disconnect the test resistor. This manual test bypasses the motion lockout; do not hold it low.
5. Timer	Start P3 near 330k (measured with power off). TIME should rise smoothly to about 3.3V, then discharge and RUN must fall. Time the interval and adjust P3 to about 60s.
6. Oscillator / LEDs	Connect LED branches. P2 about 35k gives about 0.7s per step. Confirm LED1 -> LED2 -> LED3 -> LED4 -> LED5, repeat. At timeout LED_POWER and all LED light must fall to zero.
7. PIR	Before connecting OUT, verify the module pad labels and output voltage / polarity. Connect J2, allow at least 60s to settle, then test motion in dark and light. Motion during RUN must not restart the minute.
8. Final source	Check the actual charger board terminal labels and battery charge-current specification. Use a known protected 1S, 4.2V-charge Li-ion/Li-pol cell. Confirm 5V persists for at least 10 minutes while the lighthouse is idle, then test motion. First charging tests: S1 off.

09 / Troubleshooting and limits

The one-minute RC timer and the exact module revisions require bench verification

Symptom	What to inspect
RUN stays high / timer too long	C7 leakage can prevent TIME reaching the threshold. Check polarity, clean flux, reduce P3 and retest. Measure the TIME rise. If it stalls below about 3.3V, use a verified low-leakage 100u timing capacitor or redesign the timebase; do not assume this is a pot adjustment problem.
All LEDs off while RUN high	Check Q6: collector=+5V, base through R26 to RUN, emitter=LED_POWER. LED_POWER should be around 4V. Check LED polarity and transistor pinouts.
LEDs stay on after timeout	LED anodes must be powered from LED_POWER, not +5V. Check Q6 pinout, RUN low, R27 and accidental solder bridges.
Only one LED / wrong order	Check inverted CLOCK at U4.14, U4.13 at GND, the five numbered output pins, and D2/D3 cathodes at COUNT_RESET.
Stops immediately when LEDs light	The LEDs may illuminate LDR1, causing daylight reset. Shield / reposition the LDR; adjust P1. Feedback is not fixed by simply extending the timer.
Works on USB, fails on battery	Verify 5V boost output at idle and during a run. Check battery protection and actual board revision. Auto-off at low load is not guaranteed by the retailer specification.
Turns on after power-up	PIR warmup can create a motion signal. Allow settling. The brief reset delay is not a full 60-second PIR warmup inhibit.

Limits: low-current white LEDs, indoor use, no brightness fading, no firmware, no PCB layout. No measured battery life is claimed. Simultaneous charging and operating has not been verified. The design must pass the commissioning checks before permanent installation.

10 / Sources and design checks

Component pinouts checked from manufacturer documentation; module pad order remains a physical inspection step

S1 TI TLC555, SLFS043K, Jan 2026, pin functions / mono- and astable operation
https://www.ti.com/lit/ds/symlink/tlc555.pdf

S2 TI TLC372, SLCS114F, Jun 2025, p3 pinout / p4 common-mode range
https://www.ti.com/lit/ds/symlink/tlc372.pdf

S3 TI CD4017B, SCHS027C, p1 pinout / 5V operation / reset
https://www.ti.com/lit/ds/symlink/cd4017b.pdf

S4 Semtech BC547B, GME 210-062, manufacturer datasheets linked by retailer
https://www.gme.cz/v/1489324/semtech-bc547b-ammo-bipolarni-tranzistor

S5 Semtech manufacturer datasheet mirror: default C-B-E; special pinouts can exist
https://datasheet.datasheetarchive.com/originals/library/Datasheet-024/DSA00417422.pdf

S6 GME SB00312A-1, 775-183: 2.7-12V, <0.1mA, 2s; pad order not verified
https://www.gme.cz/v/1508403/sb00312a-1-pir-modul

S7 GME LDR5516, 520-084: light 5-10k, dark 0.5M
https://www.gme.cz/v/1495847/ldr5516-fotorezistor

S8 LaskaKit LA122076 charger + 5V boost; protected battery recommended
https://www.laskakit.cz/mikro-nabijecka-boost-pro-usb-powerbank-5v--usb-c/

Checks performed: unique component references; all non-NC nets have at least two endpoints; every DIP pin is assigned or explicitly NC; safe comparator input-divider ranges; current-limiting resistors for all LEDs and transistor bases; separate switched LED supply; nominal timing and initial-stage loading calculations.

Not performed: SPICE simulation, measurements on the ordered components, verification of unlabelled module pad order, confirmation of received transistor variants, or testing with the selected battery. Update status only after recording real measurements.